Lecture 10

Robustness to Effect Model Interaction Confounding

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Today's Robustness Question

What assumption/abstraction are we perturbing?

- The specified interaction captures the relevant effect heterogeneity.

Why might this assumption fail?

- There may be unanticipated interactions between the moderator and other covariates.

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What kinds of inferential instability can result?

- Estimated effects can be sensitive to model specification
- Standard errors may understate uncertainty if interaction selection uncertainty is ignored.

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What tools exist to evaluate robustness?

- Post Double Selection (PDS) LASSO

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What are the limits of these tools?

- The theoretically motivated tuning parameter is developed only for the linear model.
- The procedure searches only for omitted moderator–covariate interactions; it does not address other forms of model misspecification.

Blackwell and Olson (2021)

Model of interest:

$$Y_i = \beta_0 + \beta_1 D_i + \beta_2 V_i + \beta_3 D_i V_i + \beta_4 \mathbf{x}_i + e_i$$

Fully Moderated model:

$$Y_i = \delta_0 + \delta_1 D_i + \delta_2 V_i + \delta_3 D_i V_i + \delta_4 \mathbf{x}_i + \delta_5 \mathbf{x}_i V_i + e_i$$

$\hat{\beta}_3$ will be an unbiased estimator of the population interaction if:

- $\delta_5 = 0$ *and/or,*
- $\gamma_v = 0$ where $D_i V_i = \gamma_0 + \gamma_v \mathbf{x}_i V_i + \varepsilon_i$

Otherwise: *omitted interaction bias*.

Why not estimate the fully moderated model?

- The number of interactions grows rapidly.
- Many interactions are likely irrelevant.
- Estimating all interactions can substantially increase variance.
- We need a principled way to identify which interactions matter.

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Enter the **LASSO**

What is the LASSO

The LASSO stands for Least Absolute Shrinkage and Selection Operator

Shrinkage estimators can reduce sampling variability and sometimes improve model fit (particularly in the presence of collinearity).

- Shrinkage estimators impose constraints on the fitted model (particularly on the size of the coefficients).
- The result of these constraints is to shrink the estimates toward zero.
- Ridge Regression and the LASSO are the two most prominent shrinkage estimators.

NB: these are *biased* estimators, so they might be good for stabilizing predictions, but their coefficients won't be particularly good for more conventional theory testing.

LASSO (the L1 norm)

The LASSO minimizes the following:

$$\sum \left(y_i - \beta_0 - \sum_j \beta_j x_{ij} \right)^2 .$$

- Doesn't necessarily use all of the variables (i.e., some coefficients could be zero)
- Main use is low-variance predictions in the presence of
  - Highly collinear variables
  - Number of parameters similar to or larger than number of observations.

LASSO / Ridge Regularization

Visualizing coefficient shrinkage in 2D

Correlation Between Variables

0.9

Make Data

Constraint Shrinkage: 1.00

0 (all shrunk) 1 (OLS)

Norm

L1 (LASSO) L2 (Ridge)

RSS Ellipse Target (% of OLS RSS)

110

≥100 draws ellipse of equal residual SS. 100 = OLS fit.

Find Tangent Point

Sets the ellipse to the exact level where it touches the constraint.

Click "Make Data" to generate data

Solution:

Post double-selection LASSO.

1. Run LASSO of Y_i on $[V_i, \mathbf{x}_i, \mathbf{x}_i V_i]$ with carefully chosen tuning parameter.
2. Run LASSO of D_i on $[V_i, \mathbf{x}_i, \mathbf{x}_i V_i]$ with carefully chosen tuning parameter.
3. Run LASSO of V_i on $[V_i, \mathbf{x}_i, \mathbf{x}_i V_i]$ with carefully chosen tuning parameter.
4. Collect all variables selected in steps 1-3 and any constitutive terms of higher order interactions into Z^*.
5. Regress Y_i on $D_i, V_i, D_i V_i, \mathbf{z}_i^*$

Properties of PDS LASSO

1. Coefficients on D_i and $D_i V_i$ are consistent
2. SE from OLS model are asymptotically correct

Choosing λ

The choice of λ is important. Previous work suggests that:

$$\lambda_{yj} = \lambda_{y0}\sqrt{\frac{1}{N}\sum_i Z_{ij}^2 \varepsilon_{yi}^2}$$

where $\hat{\varepsilon}_{yi}$ could be estimated by a first stage LASSO.

The penalty has two parts:

- an overall penalty that increases with the number of candidate variables,
- a variable-specific loading that accounts for heteroskedasticity and variable scale.

The goal is not prediction.

The goal is to avoid selecting variables because of random sample correlations.

Robustness Workflow with Inters

1. Estimate target model.
2. Estimate PDS LASSO on moderator-covariate interactions.
3. Calculate robustness quantities of interest.

inters package

The `inters` package in R estimates these models. Two examples:

1. *Measuring the Democracy Repression Nexus* (Armstrong, *Electoral Studies*, 2009)
    - Outcome: Repression (`rep`), Treatment: Behavioural Democracy (`voice`), Moderator: Institutional Democracy (`veto`), Controls: `iwar`, `cwar`, `gdp10k`, `logpop`.
2. *Protesting While Black* (Armstrong, *ASR*, 2011)
    - Outcome: Police Presence (`police1`), Treatment: African American Protest (`afam`), Moderator: Year (`evyy`), Controls: `ny`, `south`, `logpart`, `propdam`, `counterd`, `standard2`, `extcont2`, `govtarg`, `stratvar`, `viold`

Democracy and Repression

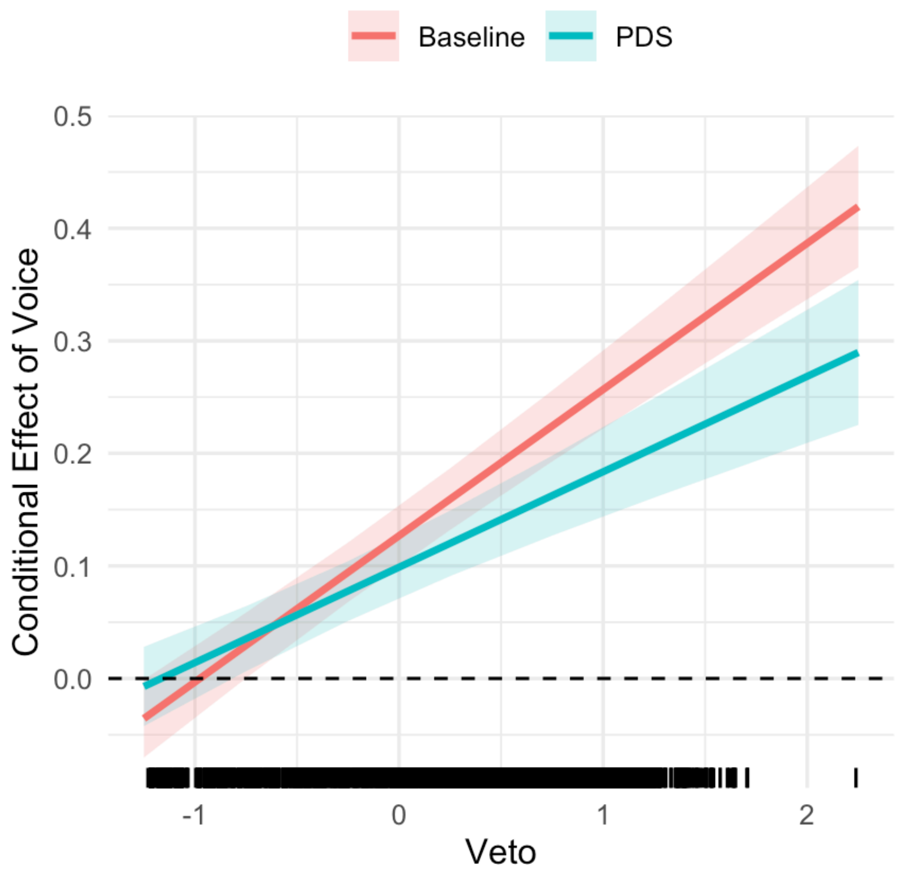
```r
library(inters)
drn <- rio::import("../module2/drn_proxies.dta")
drn_full <- lm(rep ~ voice*veto + rgdpch +
  log(pop) + iwar + cwar, data=drn)
drn_pds <- post_ds_interaction(drn,
  "voice", "veto", "rep",
  c("log(pop)", "iwar", "cwar", "rgdpch"))
drn_pds <- lm(rep ~ voice*veto + log(pop) +
  rgdpch + iwar + cwar + iwar*veto +
    rgdpch*veto, data = drn)

slp_full <- slopes(drn_full,
       datagrid(veto = seq(-1.25, 2.25, by=.5)),
       variables="voice",
       by="veto")
slp_pds <- slopes(drn_pds,
       datagrid(veto = seq(-1.25, 2.25, by=.5)),
       variables="voice",
       by="veto")
```

```r
robustness(slp_full, slp_pds) |> select(veto, estimate, base_
```

```
## # A tibble: 8 × 4
##     veto estimate base_est    ovl
##    <dbl>    <dbl>    <dbl>  <dbl>
## 1 -1.25 -0.00712  -0.0356 0.423
## 2 -0.75  0.0353    0.0294 0.842
## 3 -0.25  0.0777    0.0943 0.537
## 4  0.25  0.120     0.159  0.172
## 5  0.75  0.162     0.224  0.0700
## 6  1.25  0.205     0.289  0.0432
## 7  1.75  0.247     0.354  0.0347
## 8  2.25  0.290     0.419  0.0316
```

Effects



- Robustness statistics are not great.
- The substantive conclusion still holds - even down to the insignificance at the lowest level of veto.
- The addition of other interactions doesn't change the story.

Conclusion

Blackwell & Olson suggest that omitted interaction bias could be problematic in interaction models.

- This is protection against a particular kind of unanticipated confounding.
- The PDS Interaction model is a special case of a more general technique - Double Machine Learning (DML).